

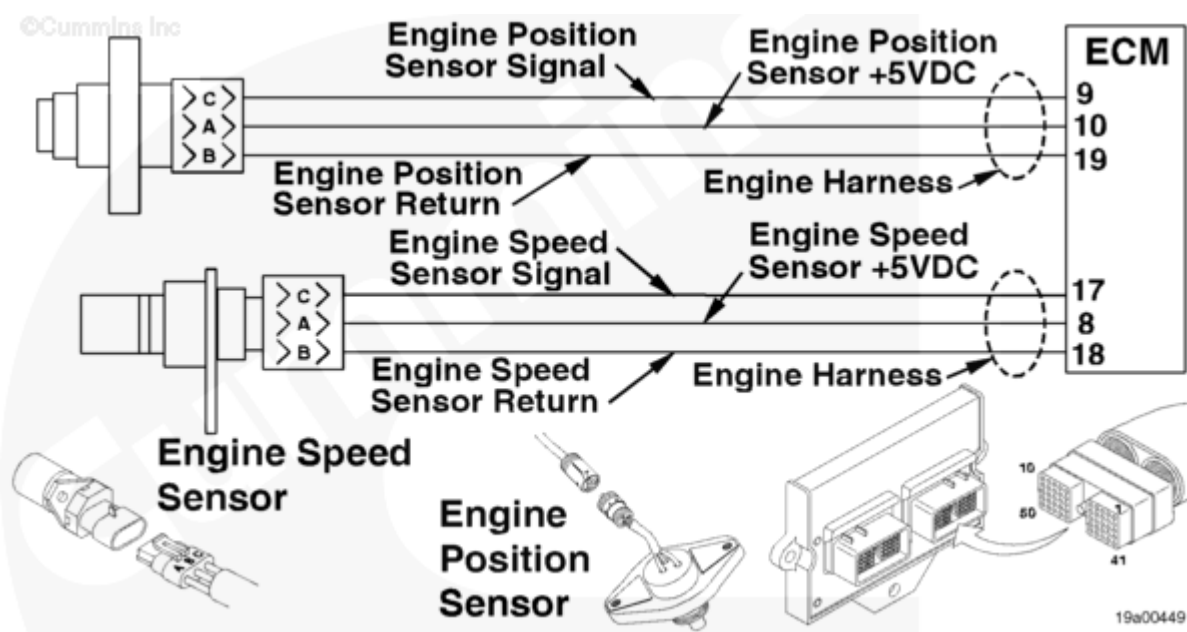
Fault Code: 121

Engine Position Sensor Circuit

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Overview

Codes	Reason	Effect
Fault Code: 121 PID(P): P190 SPN: FMI: 10 Lamp: Yellow SRT:	No engine position signal detected at pin 9 of the engine harness.	Loss of backup engine position sensor. No action is taken by the electronic control module (ECM).



Engine Position Sensor Circuit

Circuit Description

The engine position sensor provides the engine position signal to the ECM through the engine harness as well as a backup engine speed signal. The sensor **must** be powered by a positive (+) 5 VDC power source to operate. The sensor generates a signal by sensing the movement of the target holes of the flywheel. The sensor signal is 4 to 5 VDC between the holes of the flywheel, and 0 to 2.3 VDC when a hole is aligned with the sensor. A breakout cable should be used to check the engine position sensor circuit.

Component Location

The engine speed sensor and the engine position sensor are located in the flywheel housing.

Shoptalk

The engine speed sensor provides the main engine position signal to the ECM through the engine harness. The engine position sensor provides the backup engine position signal to the ECM in the event of a failed engine speed sensor. The engine speed signal is generated when the engine speed sensor detects teeth on the ring gear. The engine position signal is generated by holes in the flywheel. The ECM determines engine position by detecting a special pattern of the signal. It also determines engine speed by counting the number of voltage switches for a given period of time.

Refer to Troubleshooting Fault Code t05-121 (</qs3/pubsys2/xml/en/procedures/87/87-t05-121.html>)

Last Modified: 29-Jul-2010
